

Name: Qazi Abulaala

SaP Id: 72389

Instructor: Prof. Muhammad Imran

Question:

WRITE A C++ PROGRAM THAT TAKES INTEGER VALUES FROM THE USER AND STORES THEM IN AN ARRAY, THEN COUNTS HOW MANY EVEN NUMBERS ARE PRESENT IN THAT ARRAY. USING THIS COUNT, CREATE A SECOND ARRAY WHOSE SIZE IS EXACTLY EQUAL TO THE NUMBER OF EVEN ELEMENTS, AND COPY ALL THE EVEN NUMBERS FROM THE FIRST ARRAY INTO THIS SECOND ARRAY.

main.cpp

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Run

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5
6     int n;
7
8     // Step 1: Input size of array
9     cout << "Enter size of array: ";
10    cin >> n;
11
12    int arr[n];
13
14    // Step 2: Input elements
15    cout << "Enter elements of array:\n";
16    for (int i = 0; i < n; i++) {
17        cin >> arr[i];
18    }
19
20    // Step 3: Count even numbers
21    int count = 0;
22    for (int i = 0; i < n; i++) {
23        if (arr[i] % 2 == 0) {
24            count++;
25        }
26    }
27
28    // Step 4: Create second array of exact size
29    int evenArr[count];
30
31    // Step 5: Copy even numbers
32    int j = 0;
33    for (int i = 0; i < n; i++) {
34        if (arr[i] % 2 == 0) {
35            evenArr[j] = arr[i];
36            j++;
37        }
38    }
39
40    // Step 6: Display even array
41    cout << "\nEven Numbers Array:\n";
42    for (int i = 0; i < count; i++) {
43        cout << evenArr[i] << " ";
44    }
45
46    cout << endl;
```

Output

Enter size of array: 3
Enter elements of array:
1
2
3

Even Numbers Array:
2

=== Code Execution Successful ===